

HW04 — Data-Driven Automation Testing

Course: Software Testing / Data-Driven Automation
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1. Executive Summary

This report presents the complete results of **HW04 — Data-Driven Automation Testing** conducted on the **EShop Software Under Test (SUT)**.

- Features In Scope:**
 - FR-01 : Account Registration
 - FR-11 : Order History View (User)
 - FR-14 : Category Management (CRUD Admin)
- Data-Driven Architecture:** All test parameters, input data, and expected outcomes are decoupled into external JSON data files (registration_data.json , order_history_data.json , category_data.json).
- Test Suite Metrics:**
 - Recovered HW02 Candidates: **47 candidate test cases**
 - Excluded Cases: **1 test case** (FR-11 TC-DT-012 excluded due to backend enum state-machine constraints)
 - Approved Automated Test Suite: **46 test cases** (38 UI Web Automation + 8 API Verification/Security)
 - Assertion Pattern Coverage: **5 distinct assertion patterns** implemented
 - Multi-Browser Matrix Runs: **138 total browser executions** (46 cases × 3 browser engines: Chromium, Firefox, WebKit)
- Consolidated Multi-Browser Results:**
 - All Pass (33 cases / 99 runs):** Passed consistently across Chromium, Firefox, and WebKit.
 - All Fail (13 cases / 39 runs):** Failed consistently across Chromium, Firefox, and WebKit due to application flaws.
 - Engine-Specific Failures (0 cases / 0 runs):** No browser engine compatibility or rendering bugs observed.
- SUT Quality Assessment:**
 - Confirmed Genuine SUT Defects: 8 SUT Defects**
 - Unresolved Expected-Result Ambiguities: 0 Ambiguities remaining** (TC-DT-009 HTTP 200 OK accepted per human decision)

2. Environment Verification (ENV-01)

Environment setup was verified during Stage 0 and documented in ENV-01-environment-report.md :

Component	Target URL / Port	Technology Stack	Status
Backend API	http://localhost:3000	Node.js / Express / SQLite3	Operational
Web Frontend	http://localhost:5173	React / Vite	Operational

Component	Target URL / Port	Technology Stack	Status
Admin Panel	http://localhost:5174	React / Vite	Operational
Automation Framework	Playwright v1.62.1	Node.js ES Modules	Operational
Browser Engines	Chromium, Firefox, WebKit	Headless Execution	Operational

3. Recovered HW02 Test Cases & Traceability (Stage 1)

Test case definitions were recovered from HW02 decision tables (DT-04) and boundary value analysis (BVA-01) documents without carrying over HW02 execution pass/fail results:

Feature	Decision Table Cases (DT)	Boundary Value Analysis Cases (BVA)	Total Recovered Candidates
FR-01 (Account Registration)	13 cases (TC001 – TC013)	3 cases (BVA-TC001 – BVA-TC003)	16 candidates
FR-11 (Order History View)	12 cases (TC-DT-001 – TC-DT-012)	6 cases (TC-BVA-001 – TC-BVA-006)	18 candidates
FR-14 (Category Management)	13 cases (TC-DT-001 – TC-DT-013)	0 cases (no explicit numeric boundaries)	13 candidates
Total	38 DT cases	9 BVA cases	47 candidates

Traceability & Exclusion Note: All 47 recovered candidate cases are preserved in documentation for traceability. FR-11 TC-DT-012 is explicitly marked as excluded from automation execution in Stage 3 and was NOT executed by Playwright. It produces 0 HW04 execution evidence.

4. Data-Driven Strategy & Test Data (DATA-01)

Test data was decoupled from automation logic and stored in external JSON files under tests/<FEATURE>/data/ :

- tests/FR01/data/registration_data.json (16 records): Accommodates client-side frontend regex requiring whitespace (\s) and isolates duplicate email setup (existing_user@example.com).
- tests/FR11/data/order_history_data.json (18 records): Retains all 18 recovered HW02 records for historical traceability. **17 records are approved** for HW04 automation execution; record 18 (TC-DT-012) is marked "excluded": true and is skipped by the automation runner.
- tests/FR14/data/category_data.json (13 records): Contains parameters for category creation, deletion, empty/whitespace strings, duplicate names, 150-char strings, XSS payloads, and tags 4 API-oriented candidates (TC-DT-009 – 012).

5. Selected Automatable Test Cases & Exclusions (Stage 3)

Feature	Recovered Candidates	Approved Automated	Excluded Cases	Exclusion Justification
FR-01	16	16 cases	0	All cases fully automatable via Playwright UI.
FR-11	18	17 cases	1 (TC-DT-012)	Backend SUT API enforces enum validation (pending , confirmed , shipping , delivered , canceled). Creating an unknown status string ("processing") requires out-of-band SQLite file manipulation, violating black-box web automation boundaries.
FR-14	13	13 cases	0	9 UI cases + 4 API verification/security cases.
Total	47	46 test cases	1 test case	Required ≥36 automated cases satisfied (46 cases)

Approved Suite Breakdown

- **UI Web Automation Cases: 38 cases** (16 FR-01 + 13 FR-11 + 9 FR-14 = 38)
- **API Verification / Security Cases: 8 cases** (4 FR-11: TC-DT-003 , TC-DT-004 , TC-DT-009 , TC-DT-010 + 4 FR-14: TC-DT-009 , TC-DT-010 , TC-DT-011 , TC-DT-012 = 8)
- **Total Approved Automated Cases: 46 cases** (38 UI + 8 API = 46)

6. Generated Automation Architecture (AUTO-01)

Data-driven Playwright automation scripts were implemented in ES module format:

```
SUT_HW04/  
├─ playwright/  
│   └─ playwright.config.js      # Playwright config (Chromium, Firefox, WebKit, HTML reporter)  
└─ tests/  
    ├─ FR01/  
    │   ├─ data/registration_data.json  
    │   └─ scripts/fr01_registration.spec.js  
    ├─ FR11/  
    │   ├─ data/order_history_data.json  
    │   └─ scripts/fr11_order_history.spec.js  
    └─ FR14/  
        ├─ data/category_data.json  
        └─ scripts/fr14_category.spec.js
```

Automation Architecture Highlights

- **Dynamic Iteration & Exclusion Guard:** Scripts load external JSON at runtime using `fs.readFileSync` and iterate via `for (const tc of testCases) { if (tc.excluded) continue; test(...) } .`
- **Precondition API Helpers:** `beforeAll` hooks pre-seed required DB states (e.g. registering dedicated users and creating exact order counts/statuses via API) to prevent test-order coupling and eliminate false data-defect failures.

- **Robust Selectors:** Relies on semantic text matchers (`span:has-text(...)` , `button:has-text(...)`) and input hierarchy locators rather than brittle CSS paths.

7. Assertion Patterns & Coverage (ASSERT-01)

The test suite implements **5 distinct assertion patterns**, exceeding the minimum requirement of 3 distinct patterns:

Pattern #	Assertion Pattern Type	Mechanism / Playwright API	Code Example
Pattern 1	URL / Navigation	Web-first route matcher	<code>await expect(page).toHaveURL(/login/);</code> <code>expect(page.url()).toContain('/register');</code>
Pattern 2	Visibility / Existence	Web-first element locator	<code>await expect(errorAlert).toBeVisible();</code> <code>await expect(table).toBeVisible();</code>
Pattern 3	Element / Row Count	Async web-first count matcher	<code>await expect(rows).toHaveCount(tc.expected_row_count);</code> <code>await expect(cancelButton).toHaveCount(0);</code>
Pattern 4	Text / Dialog Content	Inner text & dialog event listener	<code>expect(errorText.length).toBeGreaterThan(0);</code> <code>expect(dialog.message()).toContain('Hủy đơn thành công!');</code>
Pattern 5	HTTP Status Code	Response status matcher	<code>expect(response.status()).toBe(401);</code> <code>expect(deleteRes.status()).toBe(404);</code>

8. Execution Baseline & Failure Classification (EXEC-01)

Single-browser execution baseline was conducted on Chromium for the 46 approved automated cases. Every failure was classified strictly per `WORKFLOW.md` criteria:

- **FR-01 Execution:** 16 Executed — 13 Pass / 3 Fail (All 3 failures grouped under Defect #1).
- **FR-11 Execution:** 17 Executed — 11 Pass / 6 Fail (1 case `TC-DT-012` excluded and not executed; Defects #2, #3).
- **FR-14 Execution:** 13 Executed — 9 Pass / 4 Fail (`TC-DT-009` Pass per human decision; Defects #4, #5, #6, #7, #8).
- **Baseline Total:** 46 Executed Automated Cases — 33 Pass / 13 Fail.

9. Multi-Browser Test Matrix & Analysis (BROWSER-01)

The 46 approved automated test cases were executed across Chromium, Firefox, and WebKit (138 total browser runs).

Consolidated Cross-Browser Execution Matrix

Feature	TC ID	Test Case Title	Chromium	Firefox	WebKit	Cross-Browser Status	Stage 6 Failure Classification / Reference
FR-01	TC001	Successful registration (all valid)	✗ FAIL	✗ FAIL	✗ FAIL	All Fail	SUT Defect (Defect #1 — Form password validation rejects !)
FR-01	TC002	Empty name	✓ PASS	✓ PASS	✓ PASS	All Pass	Form validation error alert shown
FR-01	TC003	Empty email	✓ PASS	✓ PASS	✓ PASS	All Pass	Form validation error alert shown
FR-01	TC004	Invalid email — no @	✓ PASS	✓ PASS	✓ PASS	All Pass	Form validation error alert shown
FR-01	TC005	Invalid email — no domain	✓ PASS	✓ PASS	✓ PASS	All Pass	Form validation error alert shown
FR-01	TC006	Invalid email — no local part	✓ PASS	✓ PASS	✓ PASS	All Pass	Form validation error alert shown
FR-01	TC007	Duplicate email	✓ PASS	✓ PASS	✓ PASS	All Pass	Duplicate email error alert shown
FR-01	TC008	Empty password	✓ PASS	✓ PASS	✓ PASS	All Pass	Form validation error alert shown
FR-01	TC009	Password too short (6/7 chars)	✓ PASS	✓ PASS	✓ PASS	All Pass	Length validation error alert shown
FR-01	TC010	Password missing uppercase	✓ PASS	✓ PASS	✓ PASS	All Pass	Uppercase validation error alert shown
FR-01	TC011	Password missing lowercase	✓ PASS	✓ PASS	✓ PASS	All Pass	Lowercase validation error alert shown
FR-01	TC012	Password missing digit	✓ PASS	✓ PASS	✓ PASS	All Pass	Digit validation error alert shown
FR-01	TC013	Password missing special/space	✓ PASS	✓ PASS	✓ PASS	All Pass	Special char validation error alert shown
FR-01	BVA-TC001	Password length 7 (Min – 1)	✓ PASS	✓ PASS	✓ PASS	All Pass	Length validation error alert shown
FR-01	BVA-TC002	Password length 8 (Min)	✗ FAIL	✗ FAIL	✗ FAIL	All Fail	SUT Defect (Defect #1 — Form password validation

Feature	TC ID	Test Case Title	Chromium	Firefox	WebKit	Cross-Browser Status	Stage 6 Failure Classification / Reference
							rejects !)
FR-01	BVA-TC003	Password length 9 (Min + 1)	✗ FAIL	✗ FAIL	✗ FAIL	All Fail	SUT Defect (Defect #1 — Form password validation rejects !)
FR-11	TC-DT-001	Valid user views order history	✓ PASS	✓ PASS	✓ PASS	All Pass	Table & Chờ xác nhận badge rendered
FR-11	TC-DT-002	Zero orders empty state	✓ PASS	✓ PASS	✓ PASS	All Pass	Text Bạn chưa có đơn hàng nào. visible
FR-11	TC-DT-003	Unauthenticated access	✓ PASS	✓ PASS	✓ PASS	All Pass	Text Vui lòng đăng nhập visible
FR-11	TC-DT-004	Expired/invalid JWT token	✗ FAIL	✗ FAIL	✗ FAIL	All Fail	SUT Defect (Defect #2 — API returns 403 instead of 401)
FR-11	TC-DT-005	Status confirmed view	✗ FAIL	✗ FAIL	✗ FAIL	All Fail	Status badge mismatch
FR-11	TC-DT-006	Status shipping view	✗ FAIL	✗ FAIL	✗ FAIL	All Fail	Status badge mismatch
FR-11	TC-DT-007	Status delivered view	✗ FAIL	✗ FAIL	✗ FAIL	All Fail	Status badge mismatch
FR-11	TC-DT-008	Status canceled view	✗ FAIL	✗ FAIL	✗ FAIL	All Fail	Status badge mismatch
FR-11	TC-DT-009	Cancel pending order	✓ PASS	✓ PASS	✓ PASS	All Pass	Alert Hủy đơn thành công! intercepted
FR-11	TC-DT-010	API cancel delivered order	✓ PASS	✓ PASS	✓ PASS	All Pass	API returned HTTP 400 Bad Request
FR-11	TC-DT-011	>10 orders pagination	✓ PASS	✓ PASS	✓ PASS	All Pass	Max 10 rows boundary display verified
FR-11	TC-BVA-001	0 orders (Min boundary)	✓ PASS	✓ PASS	✓ PASS	All Pass	Empty state rendered
FR-11	TC-BVA-002	1 order (Min + 1 boundary)	✓ PASS	✓ PASS	✓ PASS	All Pass	Exactly 1 table row rendered
FR-11	TC-BVA-003	5 orders (Nominal boundary)	✓ PASS	✓ PASS	✓ PASS	All Pass	Exactly 5 table rows rendered

Feature	TC ID	Test Case Title	Chromium	Firefox	WebKit	Cross-Browser Status	Stage 6 Failure Classification / Reference
FR-11	TC-BVA-004	9 orders (Max – 1 boundary)	✓ PASS	✓ PASS	✓ PASS	All Pass	Exactly 9 table rows rendered
FR-11	TC-BVA-005	10 orders (Max boundary)	✓ PASS	✓ PASS	✓ PASS	All Pass	Exactly 10 table rows rendered
FR-11	TC-BVA-006	11 orders (Max + 1 boundary)	✗ FAIL	✗ FAIL	✗ FAIL	All Fail	SUT Defect (Defect #3 — 10-order limit not enforced: 11 rows)
FR-14	TC-DT-001	Admin views categories	✓ PASS	✓ PASS	✓ PASS	All Pass	Seeded categories visible
FR-14	TC-DT-002	Add valid unique category	✓ PASS	✓ PASS	✓ PASS	All Pass	Row <i>Gia dụng</i> added to table
FR-14	TC-DT-003	Add empty/space category	✗ FAIL	✗ FAIL	✗ FAIL	All Fail	SUT Defect (Defect #4 — Empty category inserted without alert)
FR-14	TC-DT-004	Add duplicate category	✗ FAIL	✗ FAIL	✗ FAIL	All Fail	SUT Defect (Defect #5 — Duplicate category <i>Điện thoại</i> inserted)
FR-14	TC-DT-005	Add 150-char category	✓ PASS	✓ PASS	✓ PASS	All Pass	Row with 150-char string added
FR-14	TC-DT-006	Add XSS payload category	✓ PASS	✓ PASS	✓ PASS	All Pass	Plain text rendered; no script execution
FR-14	TC-DT-007	Delete category (0 products)	✓ PASS	✓ PASS	✓ PASS	All Pass	Row removed from table
FR-14	TC-DT-008	Delete category (with products)	✗ FAIL	✗ FAIL	✗ FAIL	All Fail	SUT Defect (Defect #6 — Category deleted despite active products)
FR-14	TC-DT-009	API delete non-existent ID	✓ PASS	✓ PASS	✓ PASS	All Pass	Expected status 200 OK accepted per human decision
FR-14	TC-DT-010	Unauthenticated GET API	✓ PASS	✓ PASS	✓ PASS	All Pass	API returned HTTP 200 OK
FR-14	TC-DT-011	User token POST API	✗ FAIL	✗ FAIL	✗ FAIL	All Fail	SUT Defect (Defect #7 — Non-admin token can create category)
FR-14	TC-DT-012	Invalid token GET API	✗ FAIL	✗ FAIL	✗ FAIL	All Fail	SUT Defect (Defect #8 — Invalid Bearer token)

Feature	TC ID	Test Case Title	Chromium	Firefox	WebKit	Cross-Browser Status	Stage 6 Failure Classification / Reference
							returns 200 OK)
FR-14	TC-DT-013	Category list empty state	✓ PASS	✓ PASS	✓ PASS	All Pass	Table headers rendered; 0 rows

Scoped Cross-Browser Analysis

No engine-specific differences were observed among the tested HW04 cases. All 13 failing test cases failed consistently across Chromium, Firefox, and WebKit (**All Fail** category). No rendering engine or browser-compatibility defects were observed.

10. SUT Defects & Explicit Human Decision Notes

All defect classifications are based strictly on observable UI, API response, and `api_specification.md` evidence.

10.1 Confirmed Genuine SUT Defects (8 Defects)

- Defect #1 (FR-01 — Account Registration):** Registration form validation logic rejects valid passwords containing special characters (e.g. !) with alert Mật khẩu quá yếu!... , preventing expected page redirection to /login (TC001 , BVA-TC002 , BVA-TC003).
- Defect #2 (FR-11 — Order History View):** Requesting GET /api/orders/my-orders with an invalid Bearer token returns HTTP 403 Forbidden instead of standard HTTP 401 Unauthorized (TC-DT-004).
- Defect #3 (FR-11 — Order History View):** Order history table displays all orders (e.g. 11 or 12 rows) without enforcing the stated 10-order pagination limit (TC-BVA-006).
- Defect #4 (FR-14 — Category Management):** Submitting a blank/whitespace string in the category creation form inserts an empty row into the table without displaying a validation error (TC-DT-003).
- Defect #5 (FR-14 — Category Management):** Submitting a duplicate category name (Điện thoại) inserts a duplicate category row into the table without displaying an alert Tên danh mục đã tồn tại (TC-DT-004).
- Defect #6 (FR-14 — Category Management):** Clicking "Xóa" on category Điện thoại deletes the category from the table even though active products are linked to it, instead of blocking deletion with a warning (TC-DT-008).
- Defect #7 (FR-14 — Category Management):** Submitting POST /api/categories with a regular user token (non-admin) creates a category and returns HTTP 200 OK , violating api_specification.md §6 admin authorization requirement (TC-DT-011).
- Defect #8 (FR-14 — Category Management):** Requesting GET /api/categories with an invalid Bearer token returns HTTP 200 OK instead of HTTP 401 Unauthorized (TC-DT-012).

10.2 Explicit Note on Resolved Ambiguity (TC-DT-009)

Note on TC-DT-009 (FR-14): DELETE /api/categories/99999 returns HTTP 200 OK { "message": "Category deleted" } . While standard REST conventions expect HTTP 404 Not Found for non-existent resource IDs, api_specification.md §3.4 does not explicitly document error response codes for non-existent category IDs. Per explicit human decision, the SUT's current HTTP 200 OK response is accepted as valid behavior for this test case, and TC-DT-009 is evaluated as **PASS**.

11. Audit & Process Compliance Summary (AUDIT-01)

- **Skill & Stage Integrity:** Followed `WORKFLOW.md` strictly one skill at a time (Stage 0 to Stage 8).
- **Human Approval Gates:** Stopped and obtained explicit human approval after Stage 0, Stage 1, Stage 2, Stage 3, Stage 4, Stage 5, Stage 6, and Stage 7 before continuing.
- **No Evidence Fabrication:** All execution results, screenshots (`playwright/test-results/`), error context logs, and HTML reports derive strictly from real Playwright test runs against the running SUT.
- **Black-Box Compliance:** All defect descriptions and test oracles rely exclusively on observable UI behavior, API status/payloads, and `api_specification.md` requirements. No internal SUT source code files are cited as primary defect evidence.
- **Mathematical Consistency:** 47 candidate test cases = 46 approved automated cases + 1 excluded case (`TC-DT-012`).
Multi-browser runs = 46 cases × 3 engines = 138 total runs (99 Pass / 39 Fail).